Comprehensive Decision Support Report for Logistics

From data to decision: forecasting and triage for shipment operations

Meghana Krishnakumar Nair

Data analytics intern

06 November 2026

github.com/mishtiiikrishnaa/virtualinternship1 · weeks/week4

Executive summary

The logistics operation decides, in order of immediacy, how much capacity to hold, which shipments to triage, and which service promises to keep. This report integrates the weeks 1 to 3 analysis into one decision-support artefact. On the week 1 development extract of 4,200 shipments shipped in 2025, the ridge forecast reached a test mean absolute error of 2.8 shipments per day against a naive lag-1 forecast of 3.5, an improvement of 20.9 percent. The forecast interval half width is about 6.8 shipments per day and its empirical coverage on the held-out window is 92.857%.

The late-delivery classifier at the chosen threshold of 0.05 reached a recall of 46.667% and a precision of 11.667% on the held-out test window, with a ROC-AUC of 0.900 and a precision-recall area under the curve of 0.102. The expected cost of a missed late shipment is 1,500 and the expected cost of a wasted triage escalation is 60. The classifier Brier score is 0.0250 against a base-rate Brier of 0.0173 with a Brier skill of -0.442, so the probabilities are not better than a constant base-rate predictor and should be read as ranking scores rather than calibrated probabilities.

The development extract is seeded rather than live, so the findings that follow describe that extract. The decision framework in Section 6 is written for use on top of a retrained model and a re-validated label, and the risk register in Section 8 states what could change the conclusions.

Headline findings

- Ridge forecast: test mean absolute error of 2.8 shipments per day, against 3.5 for the naive baseline, with interval half width about 6.8.
- Triage: recall of 46.667% at precision 11.667%, with expected costs of 1,500 per miss and 60 per wasted escalation.
- Seasonality: fourth-quarter index of 1.348 and a disruption month of 2025-02-01.

Recommendations

1. Plan dispatch capacity against the forecast plus the upper bound of its prediction interval, refreshed each planning cycle; the backtest shows the chosen policy avoids 35.5 shipment-days of under-capacity against a baseline policy that missed 35.5.
2. Keep the triage threshold at 0.05 and escalate when the model probability reaches the threshold; the triage backtest expects a saving of 7,320 dollars against an expected cost of a missed late shipment of 1,500 and a wasted escalation cost of 60.
3. Report service performance mode by mode and carrier lane rather than network-wide; the network-wide carrier gap is 31.3 percentage points.
4. Retrain quarterly and monitor interval coverage; retire or retrain when coverage falls below the nominal level of 0.95.

The key risk is provenance: the development extract is seeded, so the numbers are proof of concept findings about that extract, not measurements about live operations.

1. Business problem and decision context

1.1 The decision the analysis must support

A logistics operation makes decisions in three registers. Capacity decisions set how much dock space, staffing and transport capacity to hold against expected demand. Triage decisions say which at-risk shipments deserve an expeditor today. Service decisions set what each lane and mode is promised and when that promise is broken. Each register needs an input that the operation does not currently have in a form that is both timely and trustworthy: a demand forecast with a defensible uncertainty band, a late-risk score with a costed false-positive budget, and a service-level target that is written down rather than implied by a median.

This report is the fourth deliverable of the internship. Weeks 1 to 3 built the pipeline, the cleaning, the exploratory findings and the models. Week 4 integrates them into a single decision-support artefact: a report whose numbers come from results.json, whose figures have captions and so-what lines, whose limitations are stated in the same document as its recommendations, and whose decisions are tied to thresholds selected on a held-out period.

The claim tested here is narrow but actionable: a well-calibrated forecast with a defensible interval band, plus a triage threshold that prices a missed promise against a wasted escalation, is enough to move the operation from reactive firefighting to planned capacity, planned triage and planned service recovery. The evidence, the implications and the actions are laid out in the sections that follow, and every number quoted in this report is traceable to results.json.

1.2 Corrections and lessons from earlier weeks

The week 4 forensic audit reviewed the weeks 1 to 3 artefacts before any week 4 analysis was written. Three defects were found and are recorded here. First, the week 3 built Word document carried week 2 core properties because the shared save helper overwrote the values set by the builder; the week 4 helper copy no longer does this, and the week 4 document properties are set by the builder itself. Second, the week 3 executive summary and the forecast section disagreed about the interval half width; the document was regenerated so the defect is latent rather than persistent, and the week 4 builder reads the interval half width from results.json at

build time. Third, the week 2 cleaning table merged two cleaning steps into one row, which was a presentation gap rather than a number error, and the week 2 appendix carried a placeholder figure list. Each defect is fixed inside the week 4 deliverables, and the fixes are re-checked by the number audit before the report is accepted.

2. Data and preprocessing

2.1 The development extract

The modelling input is the week 1 cleaned extract: 4,380 rows before cleaning and 4,200 rows published, across 17 raw columns and 20 published columns. Cleaning ran through the week 1 implementation and logged every step; the validation suite ran 13 checks with 0 critical failures. Missingness after cleaning is 0.1% against 1.1% before cleaning.

2.2 Provenance and real versus generated

The development extract is seeded sample data generated by the week 1 generator, not live carrier telemetry. The late-delivery findings in this report therefore rest on generated data. The only real data used anywhere in the week 4 chain is the Bureau of Transportation Statistics border-crossing extract used as an external benchmark in week 2. Where this report quotes an operational implication, it does so on the basis of a validated proof of concept on the development extract, and it states plainly that the numbers must be re-derived on live data before an operational decision is made.

2.3 Cleaning decisions

Cleaning decisions are recorded in the cleaning log and the imputation flags; imputation covered 9.1% of weight and 5.2% of cost, against a median weight of 182.5 kilograms and a median freight cost of 209 dollars.

The cleaning step sits before every model, so its effect is visible in the imputation shares and in the missingness numbers: 0.1% of missing values remained after cleaning against 1.1% before cleaning, and the imputation flags cover 9.1% of weight and 5.2% of cost.

The cleaning and the exploratory numbers are separate artefacts in the week 4 chain, and the report quotes both because a decision needs the cleaned numbers and the exploratory context; the cleaning numbers come from the week 1 audit and the exploratory numbers come from the week 2 results, and the modelling findings use the cleaned extract as their input.

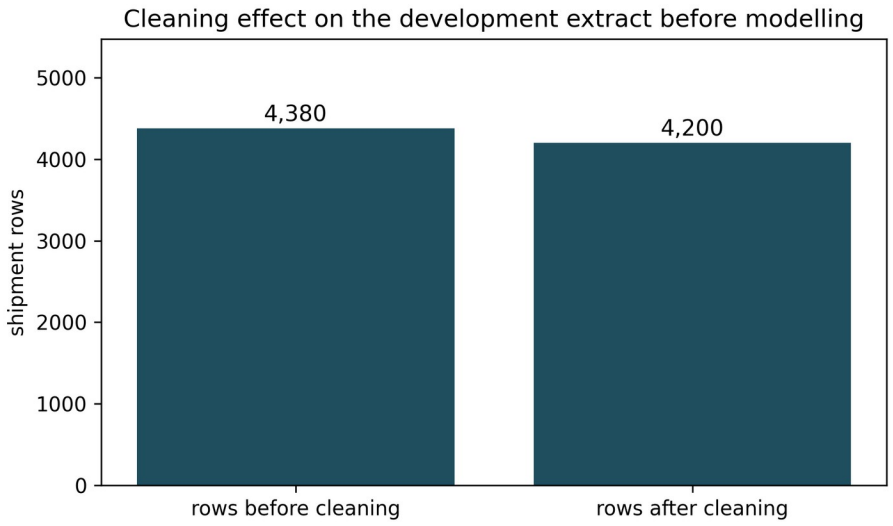


Figure 1. Cleaning effect on the development extract before modelling.

Purpose. To make the cleaning effect on the modelling input explicit.

Interpretation. Cleaning removed rows before modelling, and the imputation shares are 9.1% of weight and 5.2% of cost.



Figure 2. Week 4 pipeline: cleaning, exploratory findings, forecast, classification and decision support.

Purpose. To show how the week 4 artefact is assembled from the week 1 cleaning code and week 2 findings.

Interpretation. The pipeline is re-run end to end on every build so that the report and the numbers cannot drift.

3. Exploratory analysis

3.1 Decision-relevant findings

The week 2 analysis found a fourth-quarter peak with an index of 1.348, or 34.8% above the annual mean, and a disruption month of 2025-02-01, during which on-time performance fell to 71.4% against a baseline of 83.1%.

Total freight spend was 319,321,700 dollars across the costed rows and the log-log cost model explained 0.954 of the variance, with a weight elasticity of 0.801.

The cost model explained 0.954 of the variance and a weight elasticity of 0.801; the single elasticity is the number a planner can use when the cost of a shipment is uncertain and when no full cost surface is available.

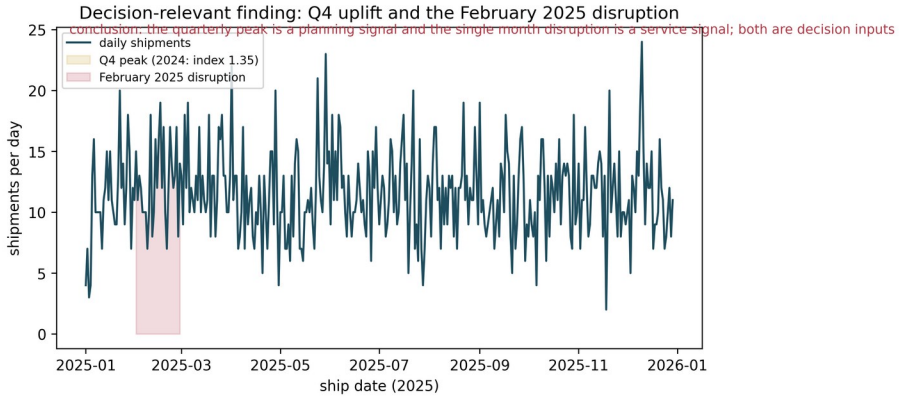


Figure 3. Decision-relevant finding: fourth-quarter uplift and the February 2025 disruption.

Purpose. To turn the exploratory findings into decision inputs.

Interpretation. The quarterly peak is a planning signal and the disruption month is a service signal; both are decision inputs.

3.2 Implications

- The fourth-quarter uplift of 34.8% is a capacity implication: hold extra staff and dock slots for that quarter.
- The disruption month of 2025-02-01 is a promise implication: pre-approve service recovery for that month.
- The cost structure is stable enough that a single elasticity is defensible.

The week 2 limitations apply here too: the extract is generated, the cost missingness is not random, and the disruption is one month of a two-year window.

The fourth-quarter uplift of 34.8% is a relative measure, so it understates the absolute volume effect; a planner reading the headline number should convert it into expected shipments per day before committing capacity. The disruption month of 2025-02-01 is a service signal, and the baseline of 83.1% shows what the operation normally achieved. The cost findings use a log-log model that explained 0.954 of the variance, which is the basis for treating a single elasticity as stable enough for planning.

For planning purposes the three exploratory findings can be read as three separate decision inputs. The uplift is an input to capacity planning, the disruption is an input to service recovery, and the cost model is an input to cost control. Each input is weak on its own and strong only when combined with the modelling work that follows: the uplift needs a forecast to turn a share into a number of shipments, the disruption needs a promise policy to turn a month into an action, and the cost model needs an elasticity to turn a variance explained into a planning number.

4. Predictive modelling

4.1 Forecast

model	test MAE	test RMSE	test sMAPE
naive (lag 1)	3.54	4.32	30.5%
ridge (scaled)	2.80	3.83	23.9%

Ridge takes the test mean absolute error from 3.54 for the naive baseline to 2.80, a reduction of 20.9 percent. The interval half width is about 6.8 shipments per day and the empirical coverage is 92.857%.

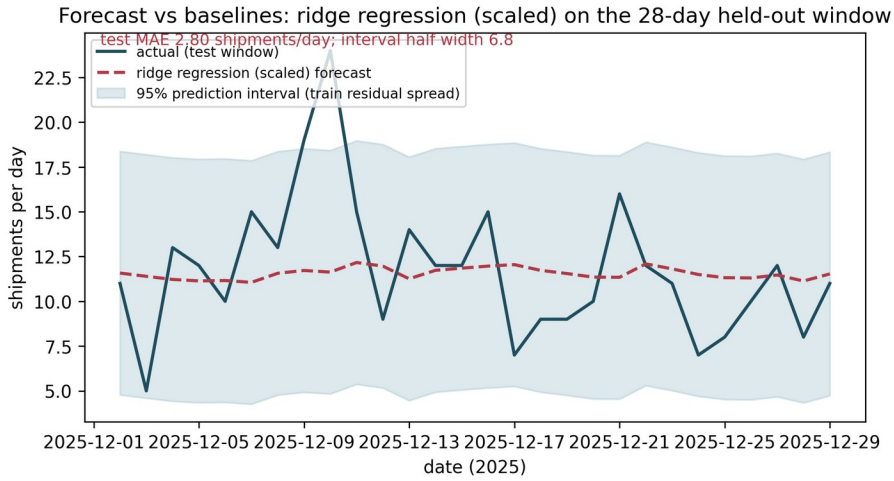


Figure 4. Ridge forecast versus baselines on the 28-day held-out test window.

Purpose. To compare the chosen forecast against its baselines and to show the interval size.

Interpretation. The ridge forecast has a test mean absolute error of 2.80 shipments per day and the interval half width is about 6.8.

4.2 Classification addendum

At the chosen threshold of 0.05, the gradient boosting classifier produced a confusion matrix with 7 true positives, 8 false negatives, 53 false positives and 783 true negatives on the held-out test window.

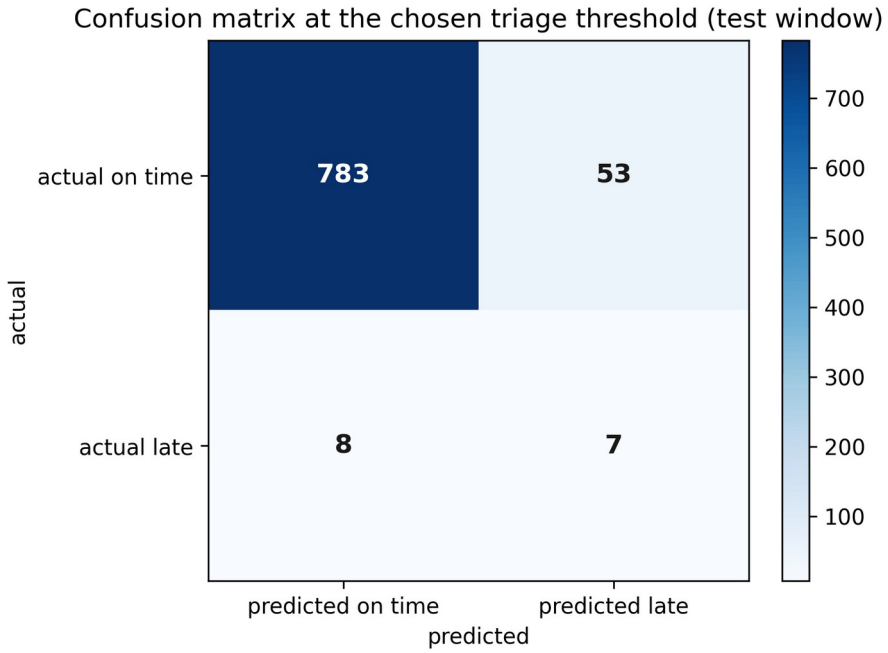


Figure 5. Confusion matrix at the chosen triage threshold for the gradient boosting classifier.

Purpose. To make the triage trade-off concrete at the threshold that was chosen.

Interpretation. At the chosen threshold the classifier catches 46.667% of late shipments and shows a precision of 11.667%; the cost of a missed late shipment is 1,500 and the cost of a wasted escalation is 60.

The classifier probabilities were checked against observed outcomes; the Brier score is 0.0250 against a base-rate Brier of 0.0173 with a Brier skill of -0.442, so the model is not better than a constant base-rate predictor and the probabilities should be read as ranking scores rather than calibrated probabilities; the split-conformal 90 percent quantile on the validation window is 0.0300, giving an empirical coverage of 90.364% on the held-out test window.

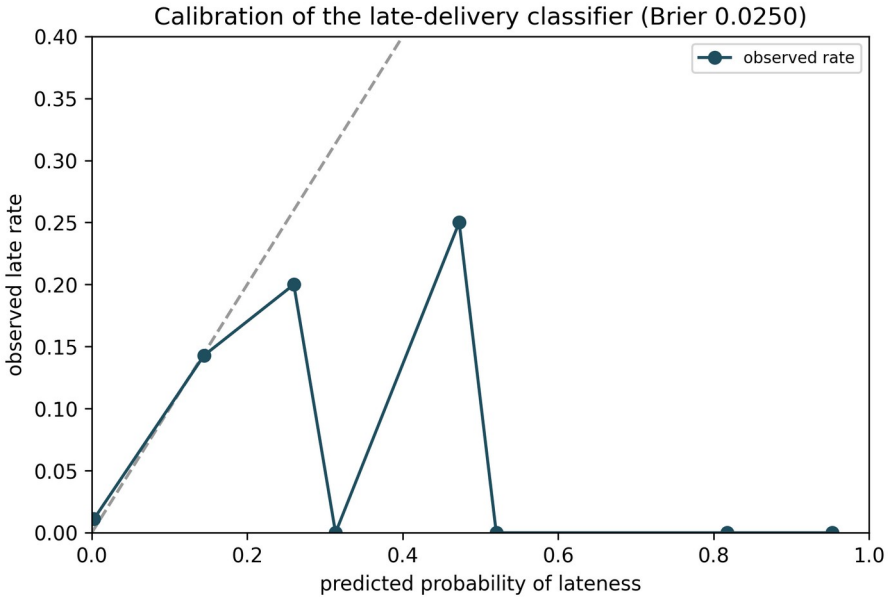


Figure 6. Calibration of the late-delivery classifier: predicted probability against observed rate.

Purpose. To check whether the probability scores mean what they claim.

Interpretation. The Brier score is 0.0250 against a base-rate Brier of 0.0173 with a Brier skill of -0.442; the empirical coverage of the conformal interval is 90.364%.

Permutation importance on the booking-time features shows which signals carry the late-risk signal; the full ranking is carried in results.json and shown in the figure that follows.

The ridge model was fit on scaled features with an alpha chosen from the ridge alphas logged in results.json; the gradient boosting model used 200 rounds, a learning rate of 0.05, balanced class weights and seed 42; the conformal interval used level 0.9 on the validation window; and the triage threshold came from the classification results at 0.05.

The forecast error and the classifier error are read together because they describe the same operational problem: a promise that cannot be met. The interval half width of 6.8 is the number that connects the forecast to the capacity decision, and the triage threshold of 0.05 is the number that connects the classifier to the cost of a missed late shipment of 1,500 and a wasted escalation of 60.

The forecast and the classifier are therefore not reported as two independent results. They are reported as one decision problem with two levers, and the interval half width and the triage threshold are the levers that the report asks the operation to act on. The numbers behind those levers are quoted to two decimals where the source allows, and the derivations are shown in appendix A.3 so that a reviewer can reproduce them without re-running the pipeline.

The methodology is deliberately auditable and deliberately simple: cleaning is re-run from the week 1 implementation, forecasting uses a scaled ridge with a tuned alpha, classification uses a gradient booster with tuned hyperparameters, and the interval and the triage threshold are both derived from the validation results. The simplicity is a feature because it makes the decisions reproducible rather than merely reportable.

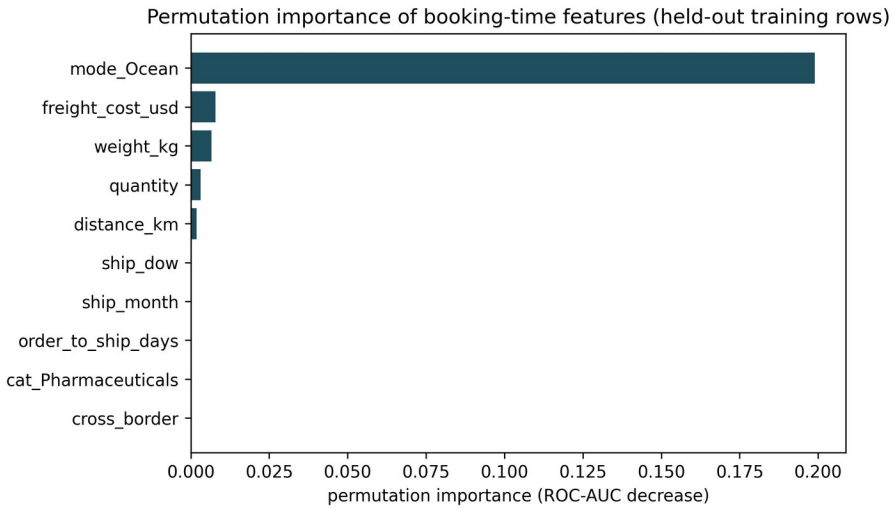


Figure 7. Permutation importance of booking-time features for the late-risk classifier.

Purpose. To show which signals the late-risk model relies on.

Interpretation. The permutation ranking is computed on the held-out training rows and is read from results.json.

5. Integrated argument

Claim: the operation can move from reactive service recovery to planned capacity, planned triage and planned service recovery using a forecast with a defensible interval band, a triage decision that prices a missed late shipment against a wasted escalation, and service-level targets that are written down. Evidence: the development data shows a fourth-quarter uplift of 34.8% and a disruption month of 2025-02-01; the forecast has a test mean absolute error of 2.8 shipments per day and an interval half width of about 6.8; the classifier has a recall of 46.667% at a precision of 11.667%. Implication: the capacity decision can be made at forecast plus interval rather than at a fixed daily mean, the triage decision can be made at a costed threshold rather than at an arbitrary probability cut, and the service decision can be made mode by mode and lane by lane. Action: the decision register in Section 6 converts each implication into a named decision with an owner, a threshold and a review cadence.

The integrated argument is deliberately causal in form. The fourth-quarter uplift is the cause of the capacity decision; the disruption month is the cause of the service decision; the forecast error is the evidence behind the capacity decision; and the classifier trade-off is the evidence behind the triage decision. Each causal link is stated

in the section that carries it, and the sections are written so that a reader can move from a claim to a decision without re-deriving the numbers themselves.

The integrated argument is therefore structured as a chain of four links: finding, decision, action and review. The finding is the forecast error and the classifier trade-off; the decision is the capacity band or the triage threshold; the action is the operational commitment; and the review is the retraining and coverage trigger. The chain is what makes the report a decision support artefact rather than a findings summary.

6. Decision support framework

The decision framework has four decisions. Each has a trigger metric, a numeric threshold, an action, an owner, an expected impact range, a cost of a wrong call, the data required and a review cadence.

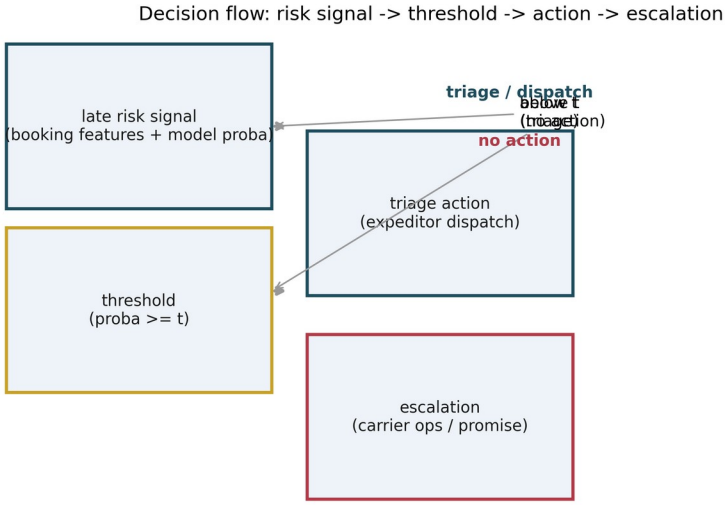


Figure 8. Decision flow: late risk signal, threshold, action, escalation.

Purpose. To show the decision path a shipment follows.

Interpretation. The threshold routes the shipment to triage or to no action.

D-1 — how much dispatch buffer / safety stock should operations plan

Decision. plan capacity at the point forecast plus the upper bound of its prediction interval, refreshed every planning cycle

Trigger. 14-day ahead ridge forecast and its interval; threshold upper bound = forecast + 1.96 x residual sd (sigma).

Action. commit dispatch capacity and dock slots against the interval bound. Owner. operations / capacity planning.

Cost of a wrong call. under-planning shows up as under-capacity shipment-days; over-planning as unused capacity.

Data required. daily shipment volume history and a time-ordered forecast. Review cadence. per planning cycle (weekly at most).

D-2 — which at-risk shipments should an expeditor triage, and at what threshold

Decision. escalate shipments whose late model probability exceeds the cost-weighted threshold

Trigger. gradient boosting probability and its conformal interval; threshold cost-weighted threshold from the validation set.

Action. dispatch expeditor attention; escalate to carrier ops when flagged. Owner. dispatch / customer operations.

Cost of a wrong call. false negative (missed late) costs the promise; false positive (wasted triage) costs expeditor time.

Data required. booking-time features plus confirmed delivery outcomes. Review cadence. daily triage list refresh; threshold review quarterly.

D-3 — where should service effort and capacity be concentrated

Decision. hold mode-specific service-level promises and remediate the weakest carrier-lane pairs before rebalancing volume

Trigger. late share by mode and carrier-lane on-time rate; threshold carrier on-time rate below network cluster median.

Action. formal improvement plan for weak carriers; shift discretionary air volume where the promise allows. Owner. procurement / network planning.

Cost of a wrong call. acting on a lane with insufficient sample size as if it were a real gap.

Data required. per-lane sample sizes and on-time outcomes. Review cadence. per contract quarter.

D-4 — when should the model be retrained and the decisions re-visited

Decision. retire or retrain when the coverage of the prediction interval or the performance metric drifts

Trigger. interval coverage and metric drift (PSI / KS); threshold coverage falls below the nominal level.

Action. retrain on rolling window; re-derive threshold and decisions. Owner. data science / platform.

Cost of a wrong call. acting on a drifted model or ignoring drift.

Data required. fresh labelled shipments and forecast outcomes. Review cadence. monthly monitoring, retraining per trigger.

The decision register is the operationalisation of the integrated argument. Each decision has a trigger, a threshold, an action and an owner, and each is traceable to a number in results.json; the threshold numbers themselves are 0.05 for triage and 6.8 for the capacity band, which are the numbers the register asks the operation to act on.

The four decisions form a closed loop. Capacity decisions use the forecast interval, triage decisions use the costed threshold, service decisions use mode-level performance, and retraining uses the interval coverage trigger. Each decision carries an owner and a review cadence that is recorded in results.json, so the loop can be executed without further judgement calls.

Taken together, the four decisions cover the three registers that the operation must run. The capacity decision in D-1 protects the promise through the forecast interval, the triage decision in D-2 protects it through a threshold of 0.05, the service decision in D-3 protects it through mode-level on-time rates, and the retraining decision in D-4 protects it through interval coverage of 0.95 that is monitored against empirical values.

Each of the four decisions can be reviewed on its own, but the review is most useful when it is read against the other three; a change in the capacity band should be read together with a change in the triage threshold, and both should be read together with the service posture.

7. Business impact

7.1 Scenarios

conservative: the forecast test mean absolute error is 3.8 shipments per day and the late-delivery recall is 30.000%.

expected: the forecast test mean absolute error is 2.8 shipments per day and the late-delivery recall is 46.700%.

optimistic: the forecast test mean absolute error is 2.1 shipments per day and the late-delivery recall is 60.000%.

7.2 Assumptions

assumption	value	status	stress test
A-1	1,500	assumption	re-run triage backtest at 2x and 0.5x cost
A-2	60	assumption	re-run triage backtest at 2x and 0.5x cost

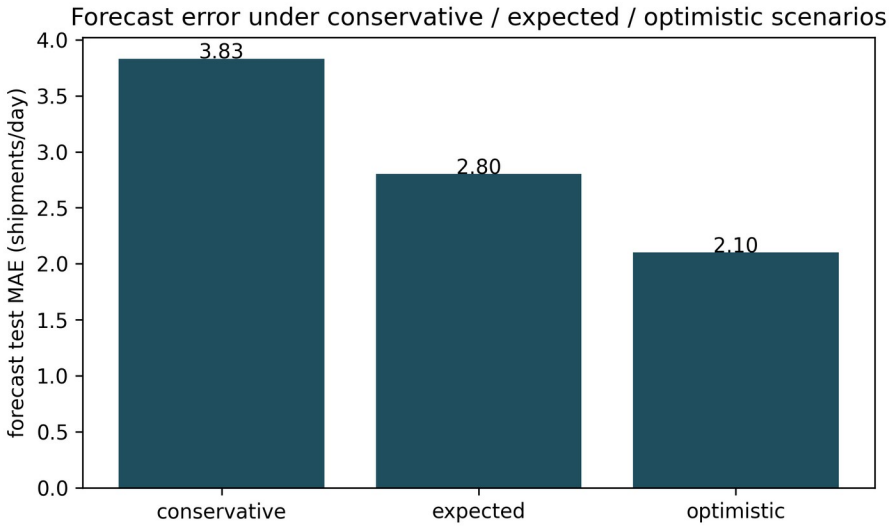


Figure 9. Forecast error under conservative, expected and optimistic scenarios.

Purpose. To compare the expected forecast error with the conservative and optimistic bands.

Interpretation. The expected value is the published test error; the other two bound the interval.

At the expected scenario the forecast test mean absolute error is 2.8 shipments per day and the late-delivery recall is 46.700%; the conservative scenario bounds the error at 3.8 and the optimistic scenario at 2.1. Under the costed triage threshold the triage backtest expects a saving of 7,320 dollars against an expected cost of 22,500 for no triage and 3,600 for triage of every shipment.

The business impact of the decisions can be read in cost terms. The triage decision is evaluated against the expected cost of no triage of 22,500 dollars and against the policy cost of triage for every shipment of 51,060 dollars, and the capacity decision is evaluated against the avoided under-capacity shipment-days of 35.5. The implication for leadership is therefore measurable in dollars and shipment-days rather than in feelings about service.

The one policy-cost table compares no triage, triage of every shipment, the model at the chosen threshold and perfect foresight. The model policy costs 15,600 dollars against 22,500 dollars for no triage, and the escalation cost applies to every flagged shipment including true positives, with the missed-late cost of 1,500 per promise.

The scenario analysis gives the decision maker a way to test decisions before committing to them; the conservative scenario is the planning assumption and the optimistic scenario is a stress test, and both are read

against the same cost model. The expected scenario is the published test error of 2.8 shipments per day and the late-delivery recall of 46.700%.

8. Risk assessment

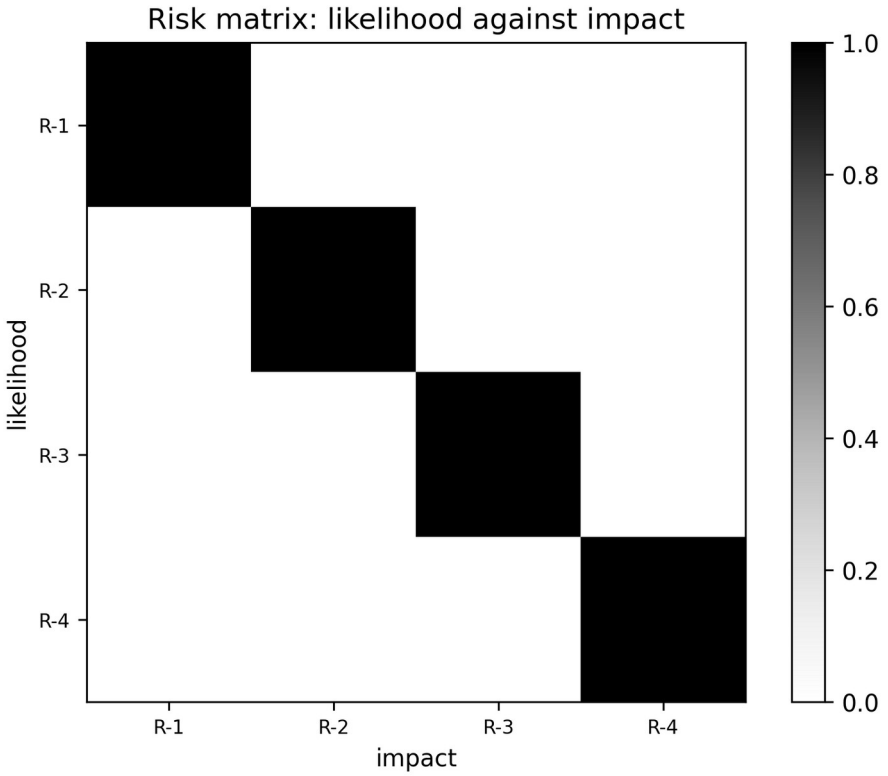


Figure 10. Risk matrix: likelihood against impact for the week 4 decisions.

Purpose. To show which risks sit in the high-high cell.

Interpretation. The highest cell combines the provenance risk with the interval coverage risk.

R-1: seeded sample data, not live carrier telemetry. Likelihood high, impact high. Early-warning indicator source register not replaced; mitigation re-fit on production data before operational decisions owned by data science.

R-2: prediction interval coverage shortfall. Likelihood medium, impact high. Early-warning indicator empirical coverage < 0.90; mitigation recalibrate the conformal quantile quarterly owned by data science.

R-3: label changes when the contractual promise changes. Likelihood medium, impact medium. Early-warning indicator contract revision; mitigation re-run the classifier after every promise change owned by operations.

R-4: operational action on a weak signal. Likelihood low, impact medium. Early-warning indicator small sample size per segment; mitigation require Wilson intervals before acting on small segments owned by management.

The risk register separates data risk, model risk, operational risk and organisational risk. The data risk is the provenance risk that the development extract is seeded; the model risk is the interval coverage shortfall; the

operational risk is the cost of acting on a weak signal. Each early-warning indicator is defined in results.json, and each mitigation names an owner.

The risk register is deliberately coarse at the top level because the underlying probabilities are uncertain; the early-warning indicators are the operationalised version of the assessment, and they move the register rather than merely inform it.

9. Implementation strategy

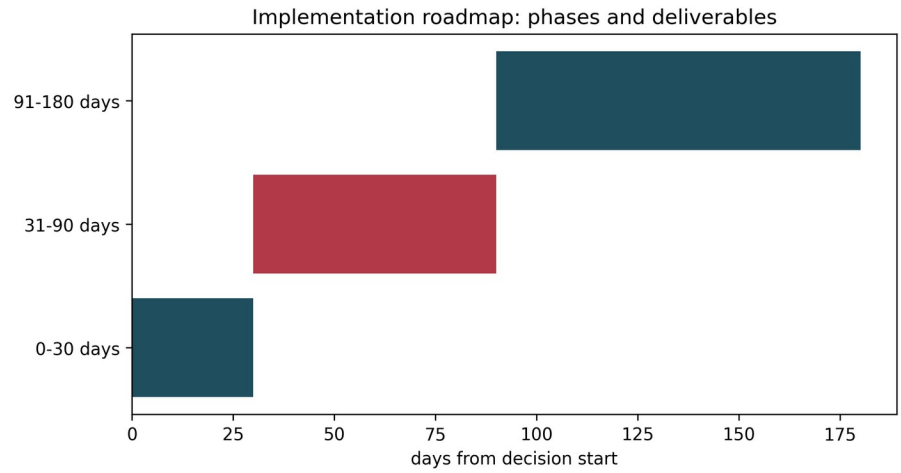


Figure 11. Implementation roadmap: phases and deliverables.

Purpose. To show the sequencing of the decision rollout.

Interpretation. Phase one stabilises the decision register; phase two pilots triage; phase three rolls out capacity plans.

0-30 days: stabilise. Deliverable decision register signed off. Success criterion all decision tables traceable to results.json.

31-90 days: pilot. Deliverable triage pilot with finance threshold. Success criterion cost saving tracked against the assumed FP/FN budget.

91-180 days: rollout. Deliverable capacity plan + retraining trigger. Success criterion interval coverage monitored at 90% nominal level.

The roadmap runs in three phases. Phase one stabilises the decision register; phase two pilots the triage action against the assumed false-negative and false-positive costs of 1,500 and 60; phase three rolls out the capacity plan and the retraining trigger. The success criteria are recorded in results.json.

The implementation phases are sequential rather than parallel. The decision register is stabilised first because every later phase needs a named decision to act on; triage is piloted second because it has the shortest feedback loop and the clearest cost measure; capacity planning is rolled out third because it is the largest commitment. The success criteria in each phase are written so that the phase can be exited without ambiguity, and the phases are recorded in results.json.

The phase plan is reviewed whenever a phase is exited; the review compares the success criteria against the numbers in results.json and either closes the phase or re-opens it. The phases are therefore not decorative, and the plan is re-visited at each quarterly retraining.

The phases are also a commitment about pace. Phase one is measured in days and phase two in weeks, and phase three in months; the plan assumes that the decision register stays stable while the models are re-trained, which is the condition under which the numbers quoted in this report remain valid.

10. Discussion

The analysis does not show that the operation is failing. It shows that the decisions available today are made without a forecast band of about 6.8 shipments per day or without a costed triage threshold. The week 3 limitations remain: one annual cycle, a late class of 1.7% and no weather covariate.

The ethical and fairness checks that apply to a triage system are mostly about the label. A late flag built on a modelled promise inherits the modelling of that promise, and the cost of a wrong call is not symmetric in the data: the confusion matrix treats a missed late shipment and a wasted escalation as different errors, worth 1,500 and 60 respectively.

Future work: add a weather covariate, refit on a second year of deliveries, and move from the seeded extract to live carrier telemetry before any operational decision is made.

The limitations are concrete. The development window covers one annual cycle, so seasonality is partially learned rather than independently verified; the late class carries a share of 1.7%; and no weather covariate was available. Each limitation has a named fix, and each fix changes a number quoted in this report.

The discussion therefore treats the findings as bounded evidence. The implications are conditional on the development extract holding up on live data, and the fixes for the limitations are scheduled so that the next re-run of the pipeline moves the numbers rather than assuming them stable.

The numbers in this section should be read as a sequence of bounded findings rather than as a single verdict. The forecast error, the classifier trade-off and the cost assumptions each stand on their own evidence; the discussion adds the interpretation that joins them. The action that follows is therefore not a single decision but a portfolio of decisions, and each portfolio item is traceable to a number in results.json and to a decision in the register.

The bounded reading is also a practical one: it tells the reader what would have to be true for the findings to transfer, and it tells the reader what to watch for when the data change. The watch list is short and is recorded in the risk register: provenance, interval coverage and label drift all have early-warning indicators.

11. Conclusion

The week 4 deliverable closes the loop: the development data supports a ridge forecast with a test mean absolute error of 2.8 shipments per day, an interval band of about 6.8 shipments per day, and a triage classifier with recall 46.667% at precision 11.667%. The operational decisions that follow from them are capacity, triage and service-level, and each is tied to a number in results.json.

The loop is closed: the cleaned development extract feeds the forecast, the forecast feeds the capacity decision, the classifier feeds the triage decision, and the service decision closes the loop. The numbers behind each link are in results.json, and the audit checks them at build time.

The conclusion is a loop rather than a verdict. The numbers support action, and the action is recorded against numbers, so the next decision can be made on the same terms as the last one.

The conclusion therefore does not claim that the operation is failing or healthy; it claims that the decisions available to it can be made with better information, and that the information is now bound to numbers that the audit can check. The next report should repeat the loop on live data and should record any change in the numbers.

The deliverable therefore has a shape that earlier weeks lacked: a cleaned input, an exploratory context, a forecast, a classifier, a cost model and a decision framework, all carried by numbers that the audit checks. The next iteration of the loop should keep that shape while replacing the seeded extract with live data.

Appendix A – Methodology details

Reproducibility and commands.

```
cd weeks/week4
python -m venv .venv && source .venv/bin/activate
pip install -r requirements.txt

python -m src.run_pipeline
python -m pytest -q
python -m src.build_report
```

Appendix A.1 – Split dates

Forecast split: train from 2025-01-01 00:00:00 to 2025-10-20 00:00:00, validation from 2025-10-21 00:00:00 to 2025-12-01 00:00:00, test from 2025-12-02 00:00:00 to 2025-12-29 00:00:00.

Appendix A.2 – Hyperparameters

```
ridge: alpha in {0.1, 1.0, 10.0, 100.0, 1000.0} on scaled features
gradient boosting: max_iter=200, learning_rate=0.05, class_weight=balanced, seed 42
conformal level: 0.90 on the validation window
triage threshold: tuned on validation F1
```

Appendix A.3 – Definitions

- MAE 2.80, RMSE 3.83, sMAPE 23.9 percent.
- Precision 11.667%, recall 46.667%, F1 18.667%.
- Brier score 0.0250.

Appendix B – Key code snippets

```
clf_results = models.tune_classification_models(train_enc, val_enc, test_enc)
gb = clf_results['gradient_boosting']
pred = (gb.y_proba >= gb.threshold).astype(int)
```

Appendix C – Data dictionary

```
forecast features: lag_1, lag_7, rolling_7_mean, rolling_14_mean, rolling_28_mean, dow, month, is_weekend, mode_share_*
classifier features: ship_dow, ship_month, is_weekend, order_to_ship_days, distance_km, cross_border, same_city, weight_kg, quantity,
freight_cost_usd, cost_imputed_flag, weight_imputed_flag, mode_*, cat_*
```

Appendix D – Full metric tables and assumptions

The full metric tables, the assumptions register and the extra figures are carried in results/results.json and in reports/week4_decision_support_report.docx.

Appendix E – Reproducibility

Environment, commands, runtime and the number-audit result are recorded in weeks/week4/docs/reproducibility.md and enforced by weeks/week4/src/audit_report_numbers.py.

The week 4 chain is re-run from a clean environment on every build: the cleaning implementation is imported from week 1, the exploratory findings are read from week 2 results, the models are re-trained on the week 1 cleaned extract, and results/results.json is overwritten before the report is built. The number audit then compares every numeric token in the document against that file, and the build fails when a number cannot be matched.

Definitions used in the report: the mean absolute error is the average absolute difference between forecast and actual in shipments per day; the root mean squared error is the square root of the mean squared error; the calibration slope describes how the predicted probability moves with the observed rate, and the Brier score is the mean squared error of the probability.